30.10.2026

Exercise 7



Goal Today: Feedback, Root Locus

Remember the course objectives

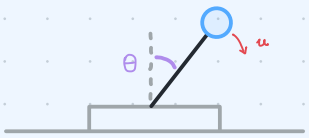
Modeling: represent real-world systems with mathematical equations.

Analysis: understand how a system behaves: input → output

Synthesis: change the system so that it behaves the way we want.

What we did so far:

W1 physical system



W2 modelling

$$ml^2\ddot{\theta} = mgl\sin\theta - c\dot{\theta} + u$$

W3 linearization

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$
$$\begin{pmatrix}\dot{\theta}\\ \ddot{\theta}\end{pmatrix} = \begin{pmatrix}0 & 1\\ 2 & -1\end{pmatrix}\begin{pmatrix}\theta\\ \dot{\theta}\end{pmatrix} + \begin{pmatrix}0\\ 0{,}2\end{pmatrix}u$$
$$y = \begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}\theta\\ \dot{\theta}\end{pmatrix}$$

W4 stability

eigenvalues $-2, +1$

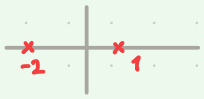
$+1 \Rightarrow$ unstable

W5 transfer function

$$G(s) = \frac{0{,}2}{s^2 + s - 2}$$

W6 poles, zeros

$$G(s) = \frac{0{,}2}{(s+2)\cdot(s-1)}$$



W7 Feedback

close the loop with a simple controller!

change the poles!

Modeling
Analysis
Synthesis

Made with Goodnotes

- The tool to understand how feedback changes our poles is called a Root Locus plot. To perfectly understand this, let's start with a recap of last week.

① Recap: transfer functions, poles, zeros

Why transfer functions?

e^{st} → [any LTI system $\dot{x}=Ax+Bu$, $y=Cx+Du$] → $G(s)\cdot e^{st}$

$$G(s) = C(sI-A)^{-1}B + D$$

- An exponential input comes out unchanged in shape, only multiplied by $G(s)$

- Every signal can be written as a sum of exponentials

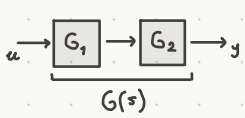
$$u(t) = \sum_k U_k \cdot e^{s_k \cdot t}$$

- $G(s)$ of a system holds the response information to every input!

incredibly powerful!

- Furthermore, derivatives and integrations become simple multiplications. Differential equations become algebra!

- We can also combine systems as if they were numbers!



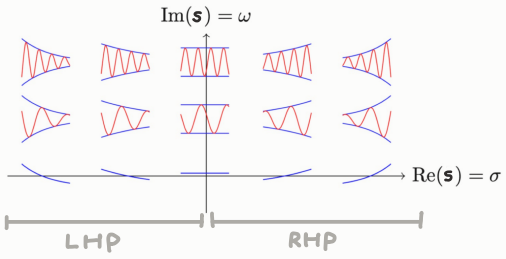
$$G(s) = G_1(s)\cdot G_2(s)$$

Made with Goodnotes

- A good way to analyze a TF is to split it to it's numerator and denominator.

$$G(s) = \frac{N(s)}{D(s)}$$

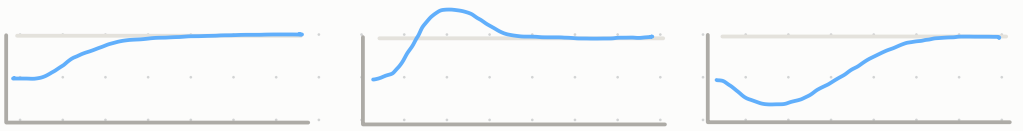
Poles:



- Poles are the roots of D(s) ($D(s)=0$), they are equal to the eigenvalues of A.
- Each pole p generates one exponential $r \cdot e^{pt}$ in the output.
- Stable $\Leftrightarrow$ all poles in LHP

Zeros:

- Zeros are the roots of N(s). ($N(s)=0$)
- They don't generate new shapes in the output. They decide how much of each shape shows up: $r \cdot e^{pt}$
- A zero in the LHP (minimum phase) adds derivative action. The system produces anticipatory behaviour.
- A zero in the RHP (non-minimum phase) adds negative derivative action. The system first goes in the wrong direction.



- Zeros never move poles. Stability is unchanged





"What if we place a zero on top of an unstable pole?
Then: $G(s) = \frac{(s-2)}{(s+5)\cdot(s-2)} = \frac{1}{s+5}$ => stable?"

"WRONG! Remember G(s) describes $u \rightarrow y$. So yes, you wouldn't see the instability in the output. But internally the system still is UNSTABLE and would blow up!"



Made with Goodnotes

② Feedback control (Recap)

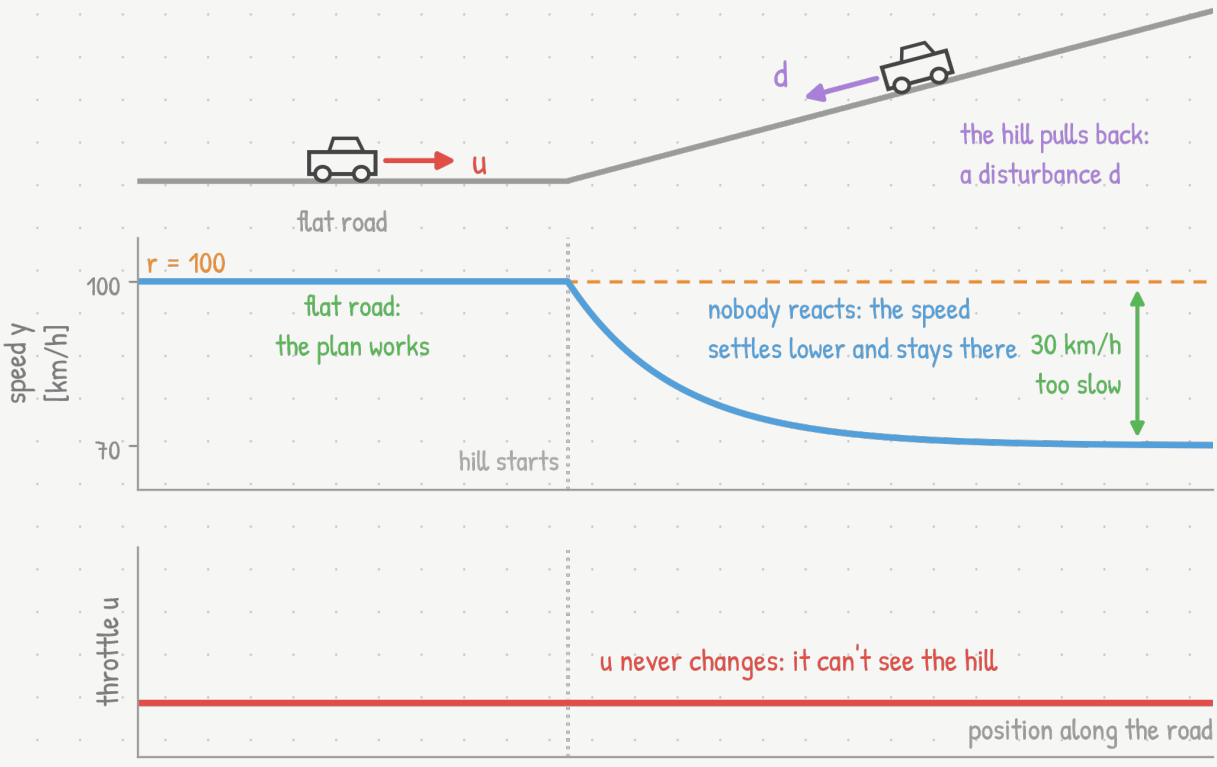
- So far we only analyzed a G that was given to us. Next, we manipulate G by adding a controller that we can design.



- A controller in front can shape the input cleverly, but like this, open-loop, it never sees the output. It never sees what it is doing.
- This method only works if the model is perfect with no disturbances.
- Let's look at this on an example:

objective: keep the speed at 100 km/h, so r = 100

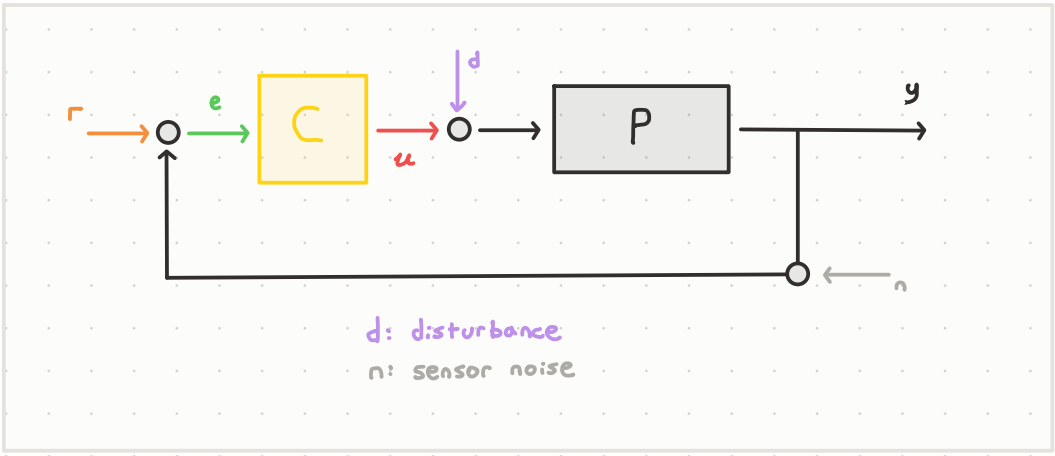
open loop: u = f(r), computed once in advance (nobody looks at the speed)



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The Fix: the controller needs to know what's going on.

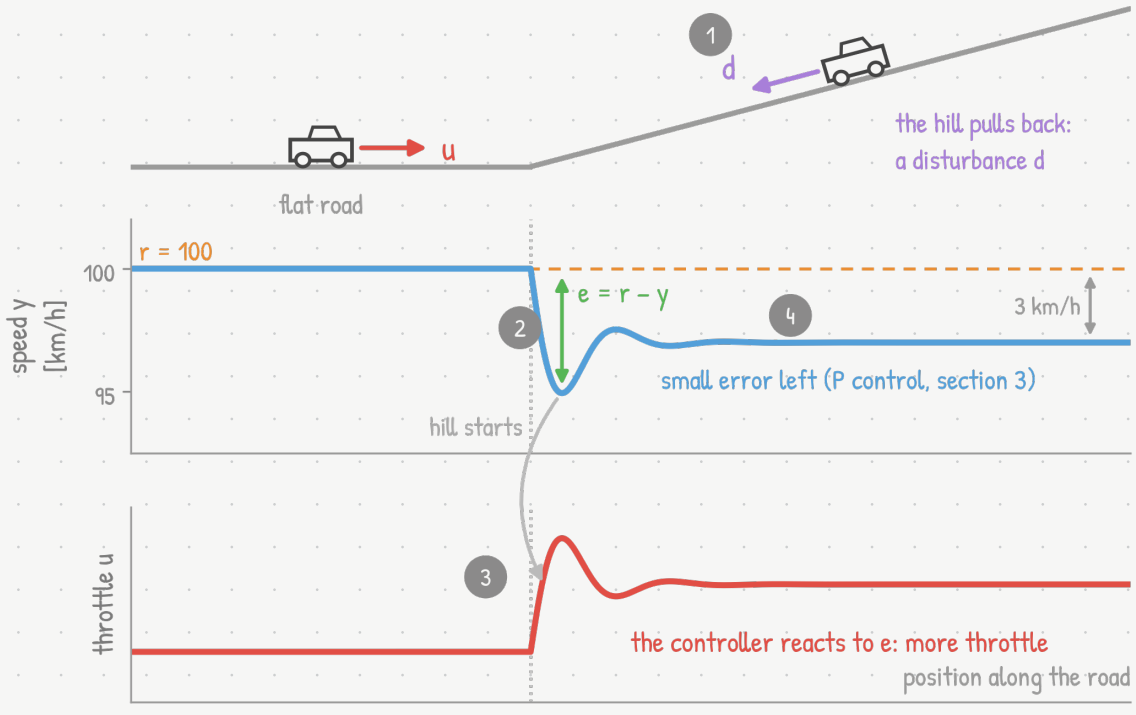
⇒ We feed the output back into the controller, and let the controller act on the error $e = r - y$



closed-loop control

objective: keep the speed at 100 km/h, so r = 100

closed loop: u = f(e), e = r − y (the controller watches the speedometer)



1 the hill slows the car 2 the error e appears 3 the controller pushes harder 4 the speed comes back

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Closed-loop transfer functions

- To inspect the CL architechture, let's use $Y(s), U(s), E(s)$ which are $y(t), u(t), e(t)$ written in terms of e^{st}. Like this, we can just multiply with the TF's to get input-output relations. ($\mathcal{L}$-Transforms)

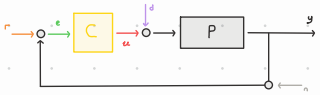
Remember: $e^{st} \rightarrow \boxed{G(s)} \rightarrow G(s) \cdot e^{st}$

- If we follow the loop:

$$\begin{aligned} Y &= P \cdot U \\ U &= C \cdot E \\ E &= R - Y \end{aligned} \quad \Rightarrow \quad Y = P \cdot C \cdot (R - Y) \quad \Rightarrow \quad Y = \boxed{\frac{CP}{CP+1}} \cdot R$$

$\Rightarrow$ We put in r and we get our output y. The transfer function that describes this new system is:

$$\boxed{T(s) = \frac{C(s) \cdot P(s)}{C(s) \cdot P(s) + 1}}$$



complementary sensitivity
closed-loop TF

$\Rightarrow$ By adding a controller and feedback we completely changed the TF of our system, along with its poles!

$\Rightarrow$ If we define our open-loop TF as $\underbrace{L(s) = C(s) \cdot P(s)}$ with $L(s) = \frac{N(s)}{D(s)}$

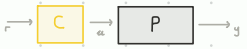


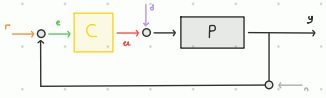
$$T = \frac{N/D}{1 + N/D} = \boxed{\frac{N(s)}{D(s) + N(s)}}$$

$\Rightarrow$ So our closed-loop poles are defined by the roots of $D(s) + N(s)$. We can manipulate this through $C(s)$!

? Your turn:

$$\underbrace{L(s) = \frac{s+1}{s^2+2s+2}} \quad \Rightarrow \quad \underbrace{T(s) =}$$





Made with Goodnotes

- We can also define $S(s) = \frac{E}{R} = \frac{1}{1 + L(s)}$, called the sensitivity.
- $T(s) + S(s) = 1$

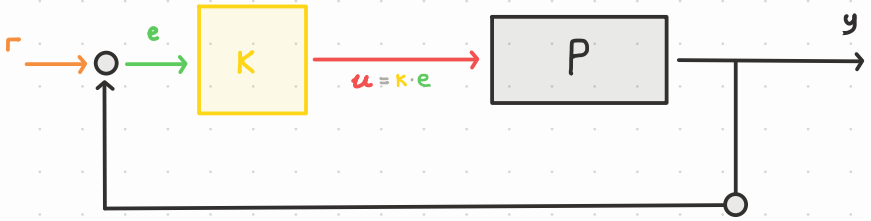
③ Proportional Feedback

- We now know how our TF gets influenced by feedback and a controller $C(s)$. Now let's start with the simplest choice for $C(s)$.
- If our error is large, we push more. If our error is small, we push less. As basic as that! So our controller is just a constant K:

$$u = K \cdot e$$

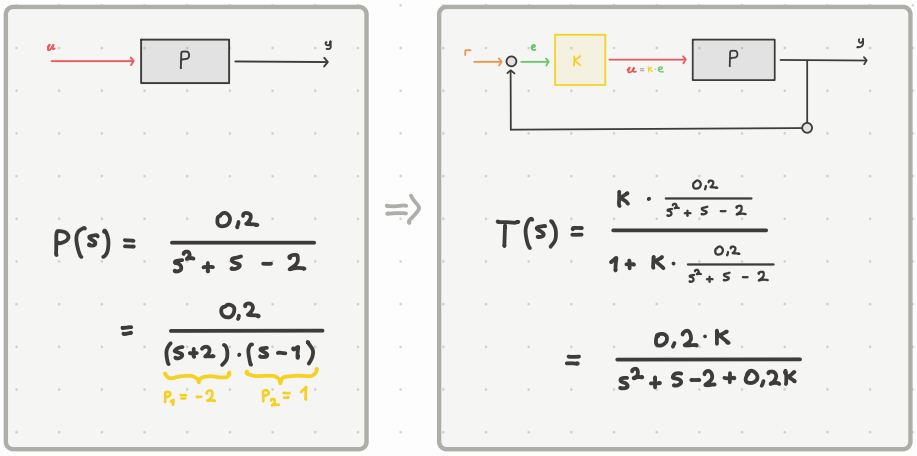
$$C(s) = K$$

K: gain

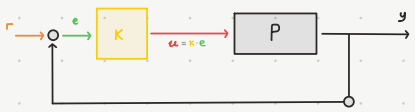


? $L(s) =$ $T(s) =$

- So for different values of K we get different closed-loop poles. We can influence our system by varying K. Let's see how this happens on an example:



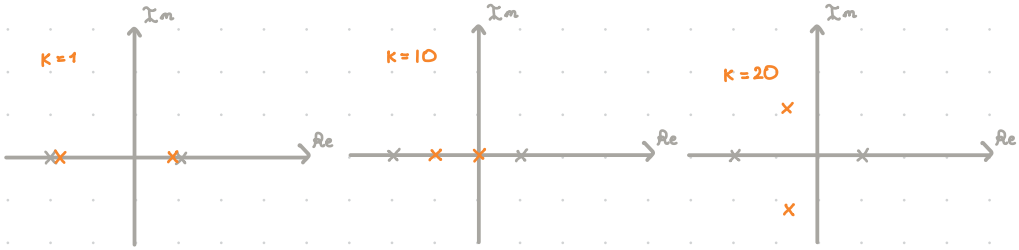
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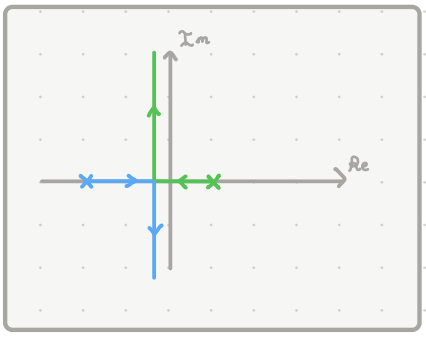
$$T(s) = \frac{0{,}2 \cdot K}{s^2 + s - 2 + 0{,}2K}$$

Solve: $s^2 + s - 2 + 0{,}2K = 0$

for different K:



⇒ We can thrace out the paths that the poles take and obtain:



We can now see that for some K both poles are in the LHP. That means we can use this simple controller to make our system stable!

The plot we just created is called root locus and it shows the location of the closed-loop poles, based on the open loop poles.

- And it allows to quickly see if a controller is feasible or not, like we just did!

- Ideally, we don't want to calculate CL pole positions for all values of K to sketch the root locus.

- Luckily, there are rules we can follow:

$P = N/D$

$$T(s) = \frac{K \cdot N(s)}{D(s) + K \cdot N(s)} \quad \Rightarrow \quad \text{CL-poles: } D(s) + K \cdot N(s) = 0$$

1. The closed-loop poles are symmetric about the real axis.

2. The number of CL poles is equal to the number of OL poles.

3. The closed-loop poles approach the open-loop poles as $K \to 0$

$D + KN \approx D$

4. If there are as many open-loop poles as open-loop zeros, then the closed-loop poles approach the open-loop zeros as $K \to \infty$ $\quad D + KN \approx KN$

If there are more poles the extra ones go to infinity.

Made with Goodnotes

Asymptotes

Contact point / Centroid of asymptotes

$$S_{com} = \frac{\sum x_{Poles} - \sum x_{Zeros}}{\#Poles - \#Zeros}$$

$$x_i \rightarrow Coordinates\ on\ the\ Real\ axis$$

Angle of asymptotes

$$\alpha_n = \frac{2n+1}{\#Poles - \#Zeros} \cdot 180°$$

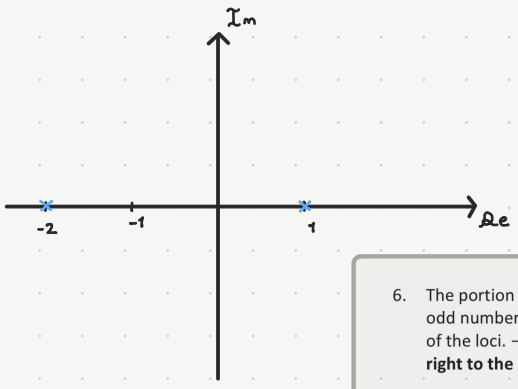
$$n = \{0; 1; \dots; (\#Poles - \#Zeros - 1)\}$$

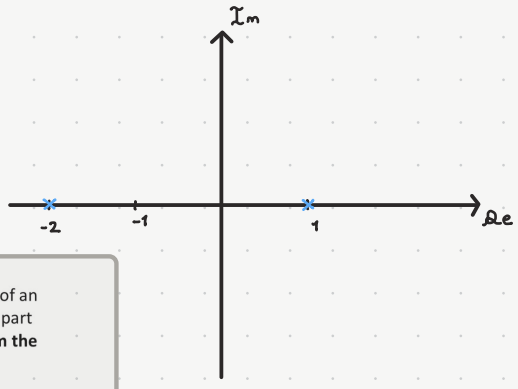
Sketching Rules

1. Root loci start at poles → go to zeros
2. There are n lines (loci) where n is the degree of Poles or Zeros (whichever is greater).
3. As k increases from 0 to ∞, the roots move from the poles of $G(s)$ to the zeros of $G(s)$.
4. When roots are complex, they occur in conjugate pairs.
5. At no time will the same root cross over its path.
6. The portion (Anteil) of the real axis to the left of an odd number of open loop poles and zeros are part of the loci. → **Roots are always sketched from the right to the left.**
7. Lines leave and enter the real axis at 90°.
8. If there are not enough poles or zeros to make a pair, then the extra lines go to / come from infinity.
9. Lines go to infinity along asymptotes.
10. If there are at least two lines to infinity, then the sum of all roots is constant.
11. K going from 0 to $-\infty$ can be drawn by reversing rule 6 and adding 180° to the asymptote angles.

Examples:

Let's start with the example from above:



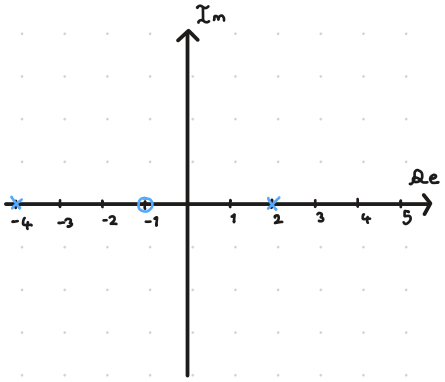


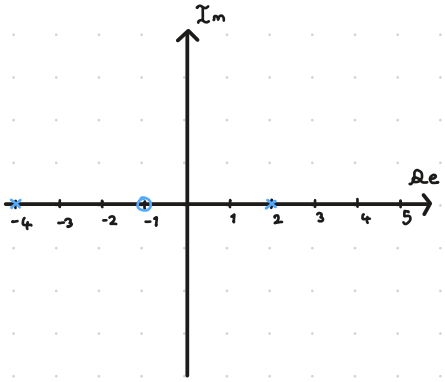
6. The portion (Anteil) of the real axis to the left of an odd number of open loop poles and zeros are part of the loci. → **Roots are always sketched from the right to the left.**

Centroid of asymp:

Angle of asymp:

Let's add zeros and look at more examples:



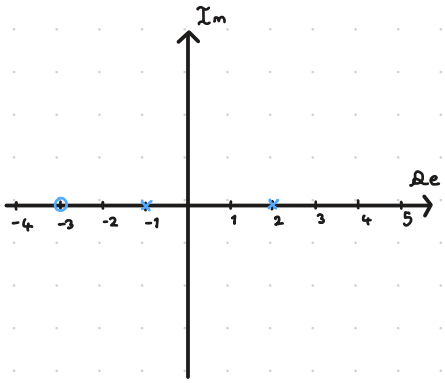


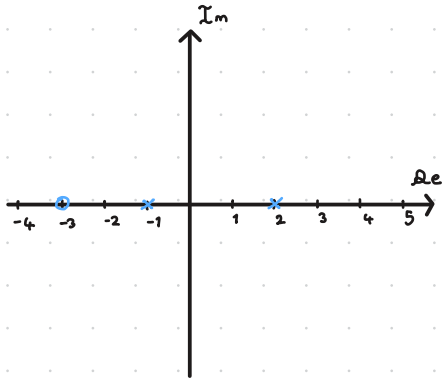
Poles: 2 # Zeros: 1

Asymp. centroid:

Asymp. angle :

Made with Goodnotes

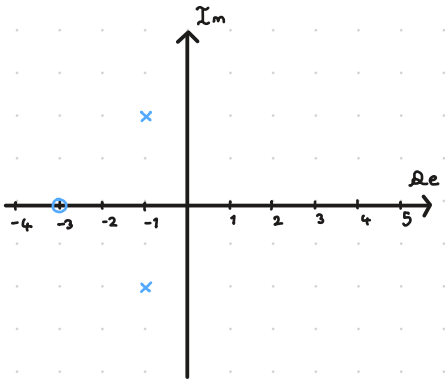


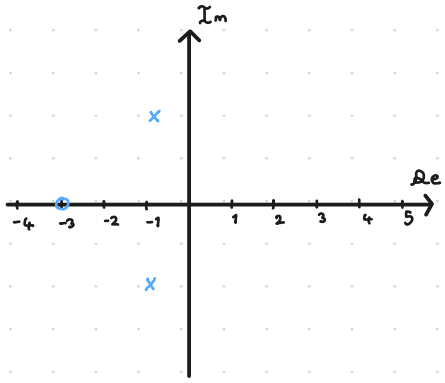


Poles: 2 # Zeros: 1

Asymp. centroid:

Asymp. angle :

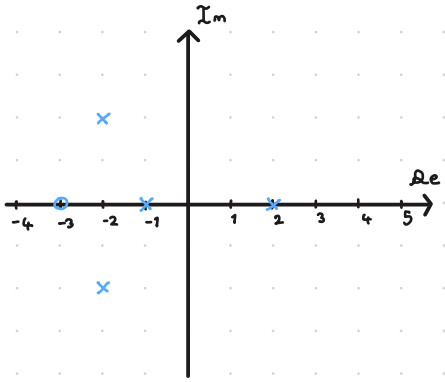


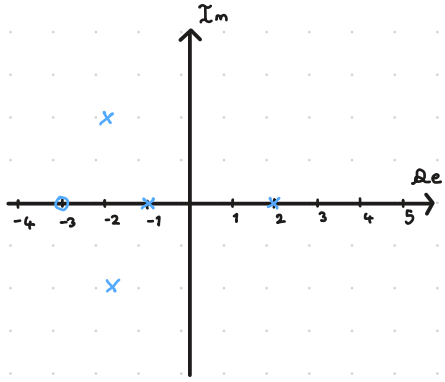


Poles: 2 # Zeros: 1

Asymp. centroid:

Asymp. angle :

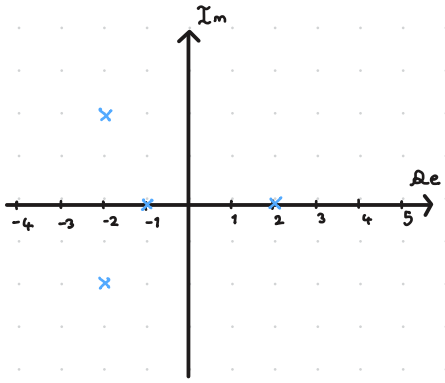


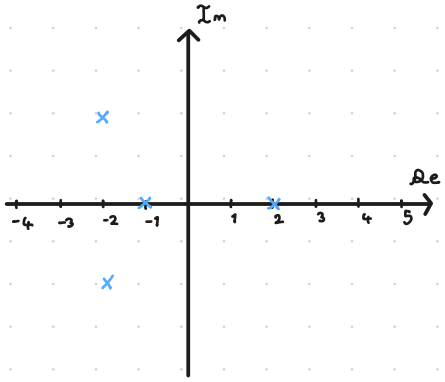


Poles: 4 # Zeros: 1

Asymp. centroid:

Asymp. angle :





Poles: 4 # Zeros: 0

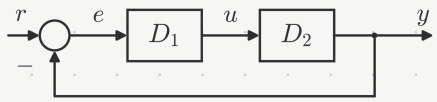
Asymp. centroid:

Asymp. angle :

Made with Goodnotes

Two things to watch:

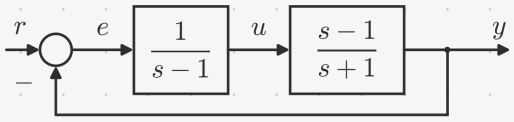
Well-posedness



$T = D_1D_2/(1 + D_1D_2)$. If $D_1D_2 = -1$

$T \to \infty$. The loop makes no sense.

Internal Stability



Unstable pole is cancelled when $G(s) = G_1 \cdot G_2$

System blows up internally.

Made with Goodnotes